

Parallel CFD Solver for Sod Shock Tube Simulation Using MPI/OpenMP with ML-Based Flow Prediction

Objectives

1. Develop a CFD solver for the one-dimensional Sod shock tube problem using compressible Euler equations and the Finite Volume Method (FVM).
2. Simulate important gas dynamic phenomena such as:
 - Shock waves
 - Expansion fans
 - Contact discontinuities
3. Implement parallelization using:
 - OpenMP
 - MPI
 - Hybrid MPI+OpenMP
4. Analyze computational performance using:
 - Execution time
 - Speedup
 - Parallel efficiency
5. Validate numerical results with analytical Sod shock tube solutions.
6. Generate CFD datasets and develop a machine learning surrogate model to predict flow variables such as pressure, density, and velocity.
7. Investigate the integration of CFD, High Performance Computing (HPC), and Machine Learning (ML) for accelerated compressible flow simulations.

Abstract

The present work focuses on the development of a parallel Computational Fluid Dynamics (CFD) solver for the one-dimensional Sod shock tube problem using the compressible Euler equations and the Finite Volume Method (FVM). The solver is designed to simulate important gas dynamic phenomena such as shock waves, rarefaction waves, and contact discontinuities. Numerical fluxes are evaluated using an approximate Riemann solver for accurate capture of discontinuities.

To improve computational efficiency, parallelization techniques based on OpenMP and MPI are implemented. OpenMP is used for shared-memory parallelism, while MPI is used for distributed-memory domain decomposition. A hybrid MPI+OpenMP approach is also investigated to enhance scalability and performance. The performance of sequential and parallel implementations is analyzed using execution time, speedup, and parallel efficiency metrics.

The numerical results, including pressure, density, velocity, and Mach number distributions, are validated against analytical solutions of the Sod shock tube problem. In addition, CFD-generated datasets are utilized to develop a machine learning surrogate model for predicting flow variables, thereby reducing computational cost for repeated simulations.

The proposed work demonstrates the integration of computational gas dynamics, high-performance computing, and machine learning for efficient and accurate compressible flow simulation applications.